

SC3.316: Mathematical Methods in Biology

Assignment 2

Due: 23 April 2026 before 11:59 pm

1. (10 points) Consider the logistic equation:

$$x_{n+1} = rx_n(1 - x_n)$$

1. (a) Explain the significance of the parameter r in the logistic map.
(b) What does x_n represent in a real-world context (e.g., population dynamics)?
 2. For $r = 2.5$ and initial value $x_0 = 0.2$:
 - (a) Compute the first 10 iterations.
 - (b) Describe the long-term behavior of the sequence.
 3. Discuss how the logistic map demonstrates the transition from order to chaos. What are the implications of this behavior in real-world systems?
2. (10 points) The Ising model is a mathematical model of ferromagnetism in statistical mechanics. Consider a 2D square lattice of size $N \times N$, where each site contains a spin $s_i = \pm 1$.

The Hamiltonian of the system is given by:

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - \mu \sum_i s_i$$

where:

- J is the interaction strength,
 - μ is the external magnetic field,
 - $\langle i, j \rangle$ denotes nearest neighbors.
1. (a) Explain the physical meaning of each term in the Hamiltonian.
(b) What is the significance of choosing $s_i = \pm 1$?
 2. (a) Describe the Metropolis algorithm used to simulate the Ising model.
(b) Explain how the energy change ΔE is calculated when flipping a spin.
 3. Write a program to simulate the 2D Ising model with periodic boundary conditions:
 - Initialize the lattice (random or all spins up).
 - Perform Monte Carlo updates using the Metropolis criterion.
 - Run the simulation for a sufficient number of Monte Carlo steps to extract the average energy.
3. (10 points) Given the linkage class partition, show that the deficiency of the network is independent of the way in which complexes are arranged within each linkage class.
4. (10 points) In molecular dynamics (MD), particles interact through interatomic potentials. One of the most widely used models is the Lennard–Jones potential, given by:

$$V(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

where:

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- r is the distance between particles,
 - ϵ is the depth of the potential well,
 - σ is the distance at which the potential is zero.
1. (a) Explain the physical meaning of the attractive and repulsive terms in the Lennard–Jones potential.
 2. (a) Describe the *Velocity Verlet algorithm* used in molecular dynamics simulations.
(b) Why is it preferred over simple integration schemes?
 3. **Implementation** Write a program to simulate the system:
 - Compute forces using the Lennard–Jones potential.
 - Integrate equations of motion using Velocity Verlet.
 - Track positions, velocities, and energies over time.